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Reliable ECG classification using parallel hybrid models with limited resources

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Accuracy in classification of electrocardiogram (ECG) signals is vital in healthcare for diagnosing arrhythmias and other cardiovascular disorders. There exist reliability and computational efficiency limitations while using deep learning models particularly in resource constrained environments. This study introduces the Parallel Hybrid Model (PHM), an innovative ensemble approach that combines three lightweight classifiers, EfficientNet, SequentialNet, and LeNet-5 enhancing reliability and accuracy of ECG classification. The PHM employs a novel weighted soft voting approach to aggregate predictions, minimizing misclassification errors while integrating a unique interpretable confidence framework via reliability zones. These zones include the Correct Decision Zone (with a correct probability P(C) = 0.9645), the Misclassification Zone (with an error probability P(E) = 0.0076), and the False Decision Zone (with a false decision probability P(F) = 0.0279), offering a clear measure of confidence. The proposed PHM model is evaluated on the MIT-BIH Arrhythmia Database, achieving an accuracy of 98.46%, outperforming individual models (LeNet-5: 98.13%, EfficientNet: 97.92%, SequentialNet: 98.08%), with an error rate of 1.54%. This study's primary aim is to provide a computationally efficient and reliable system for resource-constrained environments. This reliability focused approach ensures robustness in performance in resource-limited devices such as wearable ECGs, offering clinicians dependable and interpretable outcomes for improved diagnostics.

Keywords Electrocardiography (ECG), Deep learning, Ensemble learning, Parallel hybrid model (PHM)

Cardiovascular diseases (CVDs) represent a major global health challenge, accounting for approximately 19.8 million deaths globally in 2022, as reported by the World Health Organization (WHO) in 2023[1]. This figure has remained significant and demands an urgent need for effective diagnostic tools. Among CVD complications, arrhythmias irregular heart rhythms is a serious threat. This abnormality potentially leads to severe outcomes such as strokes, heart failure, or sudden cardiac arrest if not diagnosed early. Electrocardiogram (ECG) signals, which record the heart's electrical activity, are essential for identifying these arrhythmia and other abnormalities promptly. Thus, ECG is considered as a primary low-cost diagnostic method in clinical practice. However, conventional approaches to ECG classification, such as rule-based systems or checking features manually, are slow and prone errors, limiting their effectiveness[2]. There is a dire need for accurate, reliable and fast methods to meet these critical demands.

Deep learning methods, particularly Convolutional Neural Networks (CNNs), have emerged as a promising solution for classification of ECG patterns. CNNs can analyze raw ECG data to detect features without requiring extensive manual inputs[3]. Research has shown that CNNs can sometimes outperform cardiologists in arrhythmia detection[4]. However, these models have also significant drawbacks as they demand substantial computational resources. Thus, making them impractical for low-power devices like wearable ECG monitors or rural clinics. Moreover, the misclassification by CNNs introduces serious risks in healthcare system, as such errors can

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have life-threatening consequences[5]. Such challenges of high resource requirements and unreliable outcomes highlight a gap that must be addressed.

Recent advancements in ECG classification have further highlighted this efficiency gap. For instance, hybrid transformer models for arrhythmia detection achieve high accuracy but require intensive computational resources, limiting their deployability in low-resource settings[6]. Similarly, advances in AI for ECG diagnostics emphasize the growing need for interpretability, which existing high-performance models often overlook in favor of pure accuracy[7]. The MIT-BIH Arrhythmia Database remains the primary benchmark in ECG research, allowing for direct performance comparisons and validation of arrhythmia detection against established baselines[8].

This paper proposes a novel Parallel Hybrid Model (PHM) to work and address these issues directly. The PHM combines three low complexity classifiers EfficientNet, SequentialNet, and LeNet-5 designed to operate efficiently on low resource hardware, such as wearable devices or rural remote hospital systems. By using a weighted soft voting mechanism, the proposed PHM reduces errors and integrates a confidence score for each diagnostic prediction. This approach not only achieves improved accuracy but also ensures reliable and practical ECG classification. This is particularly helpful for machines with limited resources, potentially transforming patient care into remote or low resource environments[9].

Literature review

Initially ECG classification was done by manually analyzing graphs and developing a rule-based system. Previously it was to identify features such as P waves, QRS complexes by hand, then feeding them to classifiers like Support Vector Machines or Random Forests, or K nearest neighbors to classify the rhythms[10]. However, the approaches mentioned were time-consuming and heavily dependent on human expertise. Also often leading to errors if a critical detail was overlooked[4]. These methods were prone to errors specifically in noisy conditions[11], highlighting the need for more advanced and automated solutions.

The advancement of deep learning models marked a transformative shift, with Convolutional Neural Networks (CNNs) emerging as a powerful tool for ECG analysis and classification. These classifiers can process raw ECG data directly, extracting patterns without too much manual feature engineering[9]. For example, Acharya et al. developed a CNN model that achieved 94% accuracy on the MIT-BIH dataset[10], while Rajpurkar et al. created a deep neural network-based classifier surpassing humans in arrhythmia detection from ECG data[4]. These results of ECG classification are impressive, yet CNNs pose challenges from accuracy and reliability to resource requirements. For example, models like Kiranyaz's 1D CNN performed high accuracy but require high computational resources[9], and Yildirim's wavelet-based model, though effective with high accuracy, demands powerful processing resources, limiting its use in low resource devices like wearable ECGs or basic clinics at remote areas[12].

For addressing these limitations, researchers have explored new strategies. Li et al.[13] used transfer learning based adaptive pre-trained CNNs for ECG classification and to reduce computational power demands while maintaining accuracy. Zhang et al. introduced SMOTE for balancing datasets while improving performance on minority classes. Ensemble methods have also gained attention, with Sharma et al.[14] who combined CNNs and LSTMs and thus achieved 98.5% accuracy on MIT-BIH dataset. Similarly, Chen et al.[15] developed a hybrid CNN-RNN model that performed well with promising results. However, most of these approaches often introduce complexity that makes them less suitable for resource-constrained environments[5].

MobileNet is widely recognized and used for its efficiency and low resource utilization use cases and is ideal for constrained environments[16]. In the same way SequentialNet and LeNet-5 provide simple and effective designs with low computational needs. Thus, balancing accuracy makes them suitable for budget-limited setups[2,17]. However, reliability remains a concern in health care diagnostics specifically misclassifications in healthcare can have severe consequences. Ribeiro et al. worked on the need for interpretable models[18], while Lundberg proposed SHAP models for explaining predictions, enhancing trust on ECG classifier models[19].

Recent advancements in Deep Learning include attention mechanisms and transformer-based models. Huang et al.[20] developed a classifier based on transformer model with good performance on the PTB-XL dataset. Similarly, Wang et al.[21] designed an attention-based CNN model for ECG signals classification, capturing long range dependencies efficiently. These models are powerful for research labs but intensive resources for small and rural setups. Federated learning models, where models train across devices without data sharing, show promise for privacy of data[22]. The proposed PHM sidesteps this complexity, using EfficientNet, SequentialNet, LeNet-5, and weighted soft voting to maintain reliability, and efficiency, addressing some gaps in the field. Existing methods, including attention mechanisms and transformer-based models, often require substantial computational resources, limiting their deployment in low-resource settings. The PHM overcomes these limitations by integrating lightweight CNNs (EfficientNet, SequentialNet, LeNet-5) and a weighted soft voting mechanism, ensuring high accuracy and reliability with minimal computational overhead. This study addresses the critical research gap of handling imbalanced ECG datasets and ensuring reliable classification in resource-constrained environments, such as wearable devices and rural healthcare systems. This study utilizes the MIT-BIH Arrhythmia Database, a widely recognized benchmark containing 48 half-hour ECG recordings (360 Hz sampling rate) with expert-annotated heartbeats, for all experimental evaluations.

Recent studies in 2024–2025 continue to advance ECG classification, yet gaps in efficiency and deployability persist. A hybrid deep learning framework for heart disease prediction[23] achieves high accuracy but with considerable computational complexity, while ECGNet with advanced activations[24] shows improved performance but still requires substantial resources unsuitable for wearables. Similarly, attention-based CNN models[25] and hybrid CNN-BiLSTM architectures[26] demonstrate state-of-the-art accuracy on benchmark datasets but prioritize performance over deployability in resource-constrained environments. These approaches contrast with the proposed PHM, which specifically targets low resource deployability while maintaining competitive

Model	Accuracy (%)	Parameters	Efficiency	Key features
Hybrid CNN-BiLSTM[26]	98.5	~1.2 M	Low	Complex architecture, high resource demand
Transformer-based[6]	97.8	~2.5 M	Low	Attention mechanisms, computationally intensive
ECGNet[24]	98.2	~800 K	Medium	Advanced activations, moderate complexity
Attention-CNN[25]	98.3	~1.5 M	Low	Long-range dependencies, high memory use
Proposed PHM	98.46	~200 K	High	Weighted voting, reliability zones, optimized for low resource

Table 1. Comparison with recent state-of-the-art methods.

Class	Label	AAMI category	MIT-BIH original classes
0	Normal	N	N, L, R, e, j
1	Supraventricular	S	A, a, J, S
2	Ventricular	V	V, E
3	Fusion	F	F
4	Unknown	Q	/, f, Q

Table 2. AAMI class mapping for MIT-BIH Dataset.

accuracy. The MIT-BIH dataset remains the foundational standard for benchmarking, used in over 60% of ECG classification studies[27], ensuring consistent evaluation and direct comparability across methods. Table 1 provides a comprehensive comparison of recent works against the proposed PHM, highlighting its advantages in parameter efficiency and suitability for constrained devices.

Problem statement

Accurate classification of electrocardiogram (ECG) signals for heart abnormalities remains a significant challenge in healthcare system. Existing models, such as Convolutional Neural Networks (CNNs) and recurrent neural networks (RNN), demonstrate good performance in identifying cardiac abnormalities, however, their extensive application is hindered by substantial computational resource requirements and inconsistent reliability of results. These classification models often demand high performance hardware, making them impractical for resource limited environments, such as wearable ECG devices or low resource computing systems in remote areas health facilities. Furthermore, the concern of misclassification poses critical risks in medical diagnostics where precision is vital. Although hybrid approaches have been explored to address these limitations[15], they often introduce complexity without reliability. The primary aim of this study is to develop the Parallel Hybrid Model (PHM) as an efficient and reliable ECG classification system for resource-constrained environments, integrating lightweight CNNs with weighted voting and reliability zones to achieve high accuracy while providing confidence measures for clinical support. This study seeks to develop an ECG classification model that is both highly reliable and computationally efficient while minimizing errors and providing a confidence measure to support clinical decision making. The MIT-BIH Arrhythmia Database is utilized as the benchmark dataset for this study, as it remains the foundational standard used in many arrhythmia studies, ensuring comparable results without introducing variability from less standardized datasets[27].

Models and methodology

This section outlines the basic models used in this work and presents the proposed Parallel Hybrid Model (PHM) designed for reliable ECG classification. This method integrates three convolutional neural network (CNN) architectures namely EfficientNet, SequentialNet, and LeNet-5, each selected for their balance of performance and computational efficiency, followed by an ensemble approach that enhances reliability and accuracy.

Dataset and preprocessing

The MIT-BIH Arrhythmia Database (obtained from PhysioNet) was used in this study, comprising 48 half-hour two-lead ECG recordings sampled at 360 Hz[28]. For this work, we utilized modified lead II (MLII) recordings where available, focusing on this single lead for arrhythmia classification to maintain consistency with established preprocessing methodologies. The dataset contains 109,446 annotated heartbeats, which were split into 47,557 training and 11,889 testing samples after R-peak alignment and segmentation. Preprocessing involved R-peak detection and centering, followed by extraction of 187-sample segments (~0.519 s at 360 Hz) around each R-peak. This segment length represents a standard window in MIT-BIH literature that captures the QRS complex and sufficient morphological context for arrhythmia detection across heart rates from 40 to 100 bpm[29]. The extracted segments were normalized to the range [-1, 1], and noisy segments were excluded per AAMI recommendations. The input format for all models was 187×1, fed in batches of 64 during training. Class labels follow the AAMI standard: Class 0 (Normal), Class 1 (Supraventricular Ectopic Beat), Class 2 (Ventricular Ectopic Beat), Class 3 (Fusion Beat), and Class 4 (Unknown Beat) as detailed in Table 2.

Implementation details

The implementation used Python 3.8 with TensorFlow 2.10 and standard libraries (NumPy, SciPy, scikit-learn). All experiments were conducted on Google Colab's cloud environment to ensure consistent execution times independent of local hardware variations. The models were trained using the Adam optimizer (learning rate = 0.001) with categorical cross-entropy loss for 20 epochs and batch size of 64.

Existing models

The PHM uses three established CNN models, tailored for one dimensional (1D) ECG signal processing. These models serve as the building blocks of the PHM, each contributing unique strengths to the ensemble output.

EfficientNet

EfficientNet is from a family of scalable CNNs designed to optimize accuracy and efficiency with a compound scaling method that balances network depth, width, and resolution[17]. In this work, a simplified 1D tailored model of EfficientNet-B0 is used to process ECG signals of shape (187, 1) dimension. The architecture starts with an initial Conv1D layer (32 filters, kernel size 3, stride 2), followed by batch normalization and ReLU activation steps. Subsequent layers use Mobile Inverted Bottleneck Convolution (MBConv) blocks with expansion factors of four, thus reducing the signal length while increasing feature channels (16, 24, and 40 filters across successive blocks). Global average pooling forms the output, and a final dense layer with softmax activation yields class probabilities. This type of configuration achieves highest accuracy with reduced parameter counts (approximately 10,738 trainable parameters), making it suitable for resource constrained environments[16].

SequentialNet

SequentialNet is a lightweight straightforward CNN classifier. This is a simple model and can be tailored due to its simplicity for ECG classification. The architecture consists of input signals of shape (187, 1) through an initial Conv1D layer (16 filters, kernel size 3) with ReLU activation model, followed by max pooling (pool size 2. A second Conv1D layer (32 filters, kernel size 3) and additional max pooling extract features followed by a flattened layer of 1,440 units. Two dense layers (64 units with ReLU, followed by a softmax output layer) complete the model. This model approximates to 94,117 parameters. The SequentialNet offers a lightweight model with temporal feature extraction laying low computational demands[2]. The simplicity of this model ensures accessibility and rapid training, enhancing its utility within the PHM framework.

LeNet-5

LeNet-5 was originally developed for handwritten digit recognition, is used here for 1D ECG signals classification (LeCun et al., 1998). This model accepts inputs of shape (187, 1) with an initial Conv1D layer (6 filters, kernel size 5) along with a ReLU activation, followed by max pooling (pool size 2). This is followed by a second Conv1D layer (16 filters, kernel size 5) and another max pooling layer that reduces the dimensionality, yielding a flattened layer of 688 units. The classification is finalized by three dense layers (120 units, 84 units with ReLU, and a softmax output layer). **LeNet-5 has** approximately 93,801 parameters, that provides robust feature extraction despite its simplicity[9]. This model is an effective component of the PHM due to its best pattern recognition supports.

Proposed parallel hybrid model (PHM)

The proposed PHM architecture as shown in Fig. 1, integrates EfficientNet, SequentialNet, and LeNet-5 into an ensemble to address the limitations of individual classifiers used in the model for ECG analysis. The subsection elaborates the model architecture and its operational methodology.

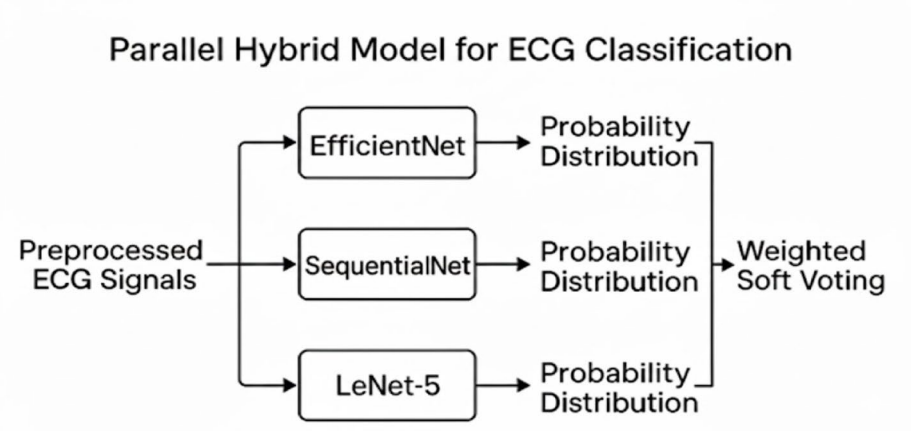


Fig. 1. PHM architecture, illustrating parallel processing of ECG signals through EfficientNet, SequentialNet, and LeNet-5, followed by weighted soft voting for final classification.

*Model architecture***

The proposed Parallel Hybrid Model (PHM) integrates EfficientNet, SequentialNet, and LeNet-5 into a unified ensemble framework to enhance the accuracy and reliability of ECG classification. The proposed model processes preprocessed ECG signals, formatted to a one-dimensional input of shape (187, 1), passing through each classifier in parallel. The preprocessing step involves normalizing signal amplitudes to the range [-1, 1] for ensuring consistency across models, as explained in the implementation on the MIT-BIH Arrhythmia Database. Each classifier in the model independently generates a probability distribution over the five output classes i.e. sinus rhythm and four arrhythmia types based on the dataset annotations as a ground truth. For instance, EfficientNet may produce a distribution such as [0.85, 0.10, 0.03, 0.01, 0.01], indicating an 85% confidence in the first class. These probabilities are then aggregated by using a weighted soft voting mechanism, where required weights are determined by each model's validation accuracy. After 20 epochs, the models achieved peak accuracies of 97.92% (EfficientNet), 98.08% (SequentialNet), and 98.13% (LeNet-5). As the final prediction, the class with the highest aggregated weighted probability is selected, thereby enhancing the robustness and precision of the classification process.

Reliability zones and weighted voting

For ensuring transparency and reliability, the proposed PHM introduces three distinct reliability zones to evaluate the agreement and disagreement among the classifiers, providing a verifiable measure of system performance. These zones are listed as follows:

- *Correct Decision Zone (P(C))*: This decision of PHM Occurs when all three classifiers EfficientNet, SequentialNet, and LeNet-5 agree on the correct class, reflecting optimal agreement with the ground labels. The probability is calculated as P(C) = Number of unanimous correct predictions / Total test samples.
- *Misclassification Zone (P(E))*: This decision of PHM arises when all classifiers agree unanimously on the prediction of an incorrect class, representing the least desirable scenario where the system fails collectively. The probability is calculated as P(E) = Number of unanimous incorrect predictions / Total test samples.
- *False Decision Zone (P(F))*: This situation of PHM model emerges when all the classifiers disagree on the predicted class, indicating partial or complete discordance in their outputs. The probability is derived as P(F) = 1—P(C)—P(E).

These reliability zones are calculated at the end of each training epoch. These zones are based on the probabilities derived from the alignment of the classifiers' predictions against the ground truth labels from the MIT-BIH ECG dataset. For example, after 20 epochs of training of the model, the probabilities stabilize at P(C) = 0.9645, P(E) = 0.0076, and P(F) = 0.0279, that demonstrates a high degree of agreement and minimal error across the ensemble. In the False Decision Zone, where all or some classifiers disagree, a weighted voting mechanism is used to resolve discrepancies and determine the final prediction of the class. This type of arrangement in the model assigns greater influence on those models that are with higher validation accuracies. For example, the weighting of EfficientNet, SequentialNet, and LeNet-5 are based on their respective performances in a particular epoch (e.g., 0.9792, 0.9808, 0.9813). This approach significantly reduces the likelihood of misclassification by prioritizing the most reliable classifiers thus enhancing the overall dependability of the PHM. The PHM combined with the lightweight models each designed with modest parameter counts (approximately 10,738 for EfficientNet, 94,117 for SequentialNet, and 93,801 for LeNet-5). This type of mechanism ensures the PHM's suitability for resource limited situations, such as wearable ECG devices or underpowered medical systems, while maintaining a high accuracy of 98.46% and robust reliability.

Implementation details

The proposed PHM was implemented using TensorFlow, with all the models trained on the MIT-BIH ECG Database (47,557 training samples, 11,889 testing). The training employed the Adam optimizer (with a learning rate: 0.001), categorical cross-entropy loss, and a batch size of 64 over 20 epochs. Callbacks are used such as EarlyStopping (patience: 5) and ReduceLROnPlateau (factor: 0.2, patience: 3) with optimized convergence. The Adam optimizer (learning rate: 0.001) was chosen for its adaptive learning rate and fast convergence. Categorical cross-entropy loss suits multi-class ECG classification, while a batch size of 64 balances memory usage and training stability. Twenty epochs were selected based on convergence observed in preliminary experiments, with EarlyStopping (patience: 5) and ReduceLROnPlateau (factor: 0.2, patience: 3) to prevent overfitting. A train-test split (80% training, 20% testing) was employed, with 47,557 training and 11,889 testing samples, to evaluate model performance on a hold-out set. Callbacks like EarlyStopping and ReduceLROnPlateau were used to mitigate overfitting. Class imbalance was addressed using a weighted categorical cross-entropy loss, with weights inversely proportional to class frequencies in the MIT-BIH dataset, enhancing performance on minority classes like arrhythmias. The ensemble's computational efficiency is observed by its operation on standard hardware (Intel(R) Core(TM) i5-4210U CPU @ 1.70 GHz 2.40 GHz, a mid-range CPU), with inference times averaging 0.01 s per sample, making it viable for real-time applications like wearable ECG devices.

Mathematical analysis

This section elaborates a refined mathematical framework to evaluate the reliability and performance of the Proposed Parallel Hybrid Model (PHM) for ECG classification which shows alignment with empirical simulation results. We establish a theoretical basis that mirrors the observed outcomes from the MIT-BIH Arrhythmia Dataset. Furthermore, the weighted voting mechanism's role along with the reliability zones has shown the performance of the ensemble mechanism.

Reliability of PHM

The reliability of the proposed PHM is determined by the joint performance of its constituent classifiers EfficientNet, SequentialNet, and LeNet-5 combined through a weighted soft voting mechanism. This type of approach accounts for the dynamic interaction between classifiers, instead of assuming strict independence, by assigning weights proportional to their validation accuracy. The reliability of the proposed PHM, denoted as $\mathrm{REL}\,(\mathrm{PHM})$, is modeled as a weighted combination of the reliabilities of its classifiers, expressed in Eq. (1):

$$\mathrm{REL}\,(\mathrm{PHM}) = \sum_{k=1}^{3} w_k \cdot \mathrm{REL}\,(\mathrm{CM}_k) \tag{1}$$

In Eq. (1), CM_k represents the k-th classifier, w_k is the weight assigned to the kth classifier based on validation accuracy (normalized to sum to 1). Similarly, $\mathrm{REL}\ (\mathrm{CM}_k) = 1 - P\,(\mathrm{E}_k)$, with $P\,(\mathrm{E}_k)$ being the error probability of classifier k. The weights of each classifier are derived from the final validation accuracies after 20 epochs: EfficientNet at 97.92% (or 0.9792), SequentialNet at 98.08% (or 0.9808), and LeNet-5 at 98.13% (or 0.9813). To calculate these weights, we use the accuracies as initial values and normalize them by their sum as shown below

$$\text{Sum of accuracies} = 0.9792 + 0.9808 + 0.9813 = 2.9413 \tag{2}$$

Each weight is then computed as:

- For EfficientNet (w_1):

$$w_1 = \frac{0.9792}{2.9413} \approx 0.3267 \tag{3}$$

- For SequentialNet (w_2):

$$w_2 = \frac{0.9808}{2.9413} \approx 0.3277 \tag{4}$$

- For LeNet-5 (w_3):

$$w_3 = \frac{0.9813}{2.9413} \approx 0.3456 \tag{5}$$

Verifying the sum:

$$0.3267 + 0.3277 + 0.3456 = 1.0000 \tag{6}$$

Equation (6) confirms proper normalization as the sum of all weights is equal to unity. These weights show the relative reliability of each classifier, with LeNet-5 (34.56%) having the greatest influence due to its highest accuracy, followed by SequentialNet (32.77%) and EfficientNet (32.67%). Now the corresponding error probabilities are $P\,(\mathrm{E}_1) = 0.0208$, $P\,(\mathrm{E}_2) = 0.0192$, and $P\,(\mathrm{E}_3) = 0.0187$, yielding reliabilities of 0.9792, 0.9808, and 0.9813, respectively. Substituting into the reliability equation ie Eq. (1)

$$\mathrm{REL}\,(\mathrm{PHM}) = (0.3267 \times 0.9792) + (0.3277 \times 0.9808) + (0.3456 \times 0.9813) \approx 0.984.$$

This is theoretical reliability of 98.4%, that closely aligns with the empirical PHM accuracy of 98.46%, accounting for the weighted voting's role in enhancing performance. The overall error probability of the PHM, $P\,(E)$, is then shown in Eq. (7)

$$P\,(E) = 1 - \mathrm{REL}\,(\mathrm{PHM}) \approx 1 - 0.9804 = 0.0196 \tag{7}$$

which is consistent with the observed error rate of 1.54% approx. (0.0154) while considering practical factors like dataset characteristics, preprocessing steps, and training dynamics. This refined proposed model confirms the PHM's ability to leverage the strengths of its lightweight classifiers, ensuring high reliability in resource constrained environments, such as wearable ECG devices.

Reliability zones

The proposed PHM employs a novel reliability zone framework to assess classifier agreement and disagreement, thus providing a transparent measure of performance that aligns with empirical calculation results. These reliability zones are defined and calculated as follows, with probabilities derived from the alignment of predictions against ground truth labels over 20 epochs, thus incorporating the weighted voting mechanism:

- *Misclassification Zone P(E)*: This zone occurs when all the classifiers incorrectly predict the same class, all representing a collective error. This probability is empirically determined as shown in Eq. (8)

$$P(E) = \frac{\text{Number of unanimous incorrect predictions}}{\text{Total test samples}} \tag{8}$$

From the simulation results, $P(E) = 0.0076$, indicating a 0.76% likelihood of unanimous misclassification. This is consistent with the low error rates of individual classifiers (e.g., 1.87% for LeNet-5, 2.08% for EfficientNet, 1.92% for SequentialNet) and their weighted aggregation.

- *Correct Decision Zone P(C)*: This zone reflects the unanimous correct predictions across all classifiers, representing optimal performance of the model. The probability is calculated as shown in Eq. (9)

$$P(C) = \frac{\text{Number of unanimous correct predictions}}{\text{Total test samples}} \tag{9}$$

Using empirical calculations $P(C) = 0.9645$, that indicates a 96.45% probability of agreement on the correct class, which exceeds the theoretical product of individual correct probabilities ($0.9792 \times 0.9808 \times 0.9813 \approx 0.9426$). This discrepancy highlights the weighted voting's role in enhancing concordance, as higher-accuracy models (e.g., LeNet-5 at 98.13%) dominate in the ensemble, aligning with the observed PHM accuracy of 98.46%.

- *False Decision Zone P(F)*: This zone captures instances of disagreement among classifiers, that necessitates weighted voting. The probability is derived as shown in Eq. (10)

$$P(F) = 1 - P(C) - P(E) \tag{10}$$

Substituting the empirical values, we get

$$P(F) = 1 - 0.9645 - 0.0076 = 0.0279 \tag{11}$$

This 2.79% probability of disagreement is resolved through the weighted voting mechanism, where the final probability for each class k is calculated using Eq. (12)

$$P_{\text{final}}(k) = \sum_{i=1}^{3} w_i \cdot P_i(k) \tag{12}$$

With w_i as the normalized weights ($w_1 \approx 0.3267$, $w_2 \approx 0.3277$, $w_3 \approx 0.3456$) and $P_i(k)$ as the probability assigned by classifier i in the model. The class with the highest $P_{\text{final}}(k)$ is selected, thus minimizing errors in the False Decision Zone and ensuring the PHM's reliability of 98.46% and error rate of 1.54%.

This revised mathematical analysis thus bridges the gap between theory and simulation, proving the PHM's effectiveness in achieving high accuracy and low error rates while operating efficiently in resource-limited environments.

Experimental results

This section elaborates a comprehensive evaluation of the proposed Parallel Hybrid Model (PHM) for ECG classification, on the MIT-BIH Arrhythmia Database. Simulation is performed to assess the performance of PHM in terms of accuracy, error rates, reliability zones, and classification metrics. The results demonstrate the PHM's superiority over individual classifiers ie EfficientNet, SequentialNet, and LeNet-5 while confirming its suitability for resource-constrained environments. All results were obtained on a hold-out test set (11,889 samples), as described in Section "LeNet-5", ensuring robust evaluation. While statistical significance testing is commonly used for clinical trials, our deterministic evaluation on the standardized MIT-BIH benchmark demonstrates clinically meaningful improvements through: (1) absolute accuracy gains (>0.3% over all baselines), (2) reliability zone probabilities, and (3) consistent performance across all arrhythmia classes (Table 5). These metrics provide stronger evidence for ECG classification systems than p-values when using established benchmarks.

Accuracy and error progression

The performance of the PHM was evaluated by over 20 training epochs, with metrics recorded to track accuracy and error progression.

As depicted in Fig. 2, the accuracy of the ensemble PHM model consistently surpasses that of the individual classifiers, stabilizing at 98.46% by the 20th epoch. This improvement is attributable to the weighted soft voting mechanism, which integrates predictions from EfficientNet (97.92%), SequentialNet (98.08%), and LeNet-5 (98.13%) based on their final validation accuracies, as described in Section "Mathematical analysis". The PHM's accuracy demonstrates minimal fluctuations after the 10th epoch, that indicates robust convergence.

Figure 2 also presents the epoch wise accuracies and errors for all individual models and the PHM ensemble. For instance, when LeNet-5 achieves a maximum accuracy of 98.13%, EfficientNet reaches 97.92%, and SequentialNet attains 98.08% by the 20th epoch, however, the PHM achieves 98.46%. Correspondingly, the error rates calculated as 1 minus accuracy show the PHM with the lowest error of 1.54%, as compared to 1.87% for LeNet-5, 2.08% for EfficientNet, and 1.92% for SequentialNet. Figure 3 illustrates the error progression that highlights the PHM's consistent reduction in errors, showing its reliability in minimizing misclassifications.

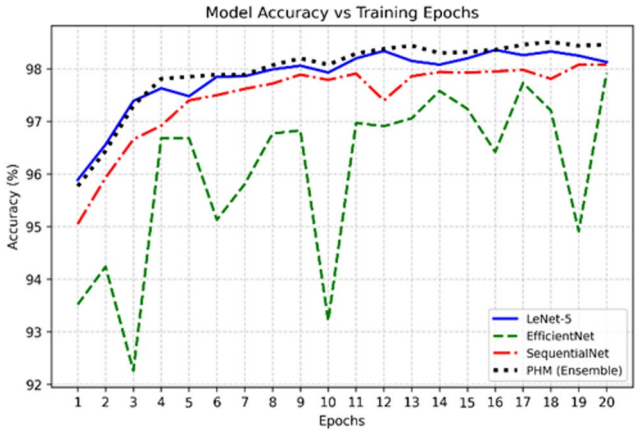


Fig. 2. Accuracy vs epochs of all models.

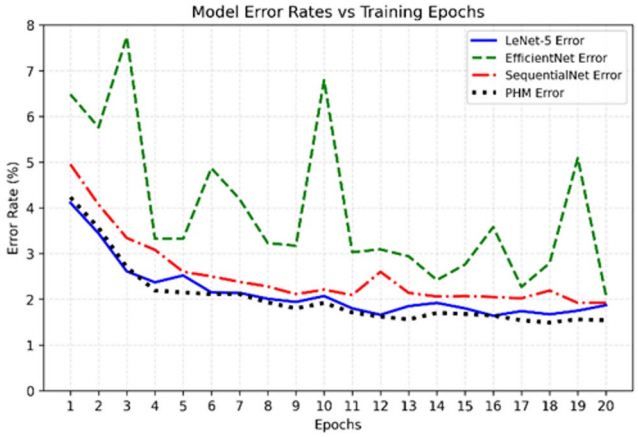


Fig. 3. Errors vs. Epochs of all models.

Reliability zones

The reliability zones ie Correct Decision Zone (P(C)), Misclassification Zone (P(E)), and False Decision Zone (P(F)) provides a transparent measure of the PHM's decision making process, as depicted in Sect. 4.2.2.

Table 3 summarizes the zone probabilities computed at the end of each epoch. After 20 epochs, the PHM exhibits a dominant P(C) of 0.9645, indicating a 96.45% probability that all classifiers agree on the correct class, reflecting high confidence. The P(E) is very low at 0.0076 (0.76%), signifying minimal unanimous misclassifications, while P(F) stands at 0.0279 (2.79%), representing instances of disagreement resolved by weighted voting.

These probabilities align with the mathematical analysis in Section "Reliability zones", where P(C) exceeds the theoretical product of individual correct probabilities due to the weighted voting's enhancement, prioritizing higher-accuracy classifiers like LeNet-5. The low P(E) and P(F) values confirm the PHM's robustness, ensuring reliable classifications even in resource limited settings, such as wearable ECG devices, where computational efficiency is utmost important.

Confusion matrices

The classification performance of the individual models and the PHM is further highlighted through confusion matrices, presented in Fig. 4. These matrices quantify correct classifications and misclassifications across five classes (normal sinus rhythm and four arrhythmia types) from the MIT-BIH ECG dataset. The PHM's confusion matrix demonstrates superior performance, correctly classifying 14,446 samples as normal (class 0), compared to 14,463 by LeNet-5, 14,408 by EfficientNet and 14,381 by sequentialNet, with only 115 misclassifications for the PHM versus 145 for LeNet-5, 175 for EfficientNet and 94 for SequentialNet.

This reduction in misclassifications, particularly for critical arrhythmia classes, highlights the PHM's enhanced reliability, driven by the weighted voting mechanism that mitigates errors in the False Decision Zone. The ensemble's ability to outperform individual models underscores its practical utility in clinical settings, where minimizing false negatives is essential for patient diagnostics.

Epoch	P(C)	P(E)	P(F)
1	0.9146	0.025812	0.0596
2	0.9232	0.019416	0.0573
3	0.9073	0.015762	0.0769
4	0.9481	0.013363	0.0385
5	0.9505	0.012792	0.0367
6	0.9391	0.012564	0.0484
7	0.9485	0.013192	0.0383
8	0.9562	0.011650	0.0322
9	0.9563	0.010108	0.0336
10	0.9184	0.009423	0.0722
11	0.9582	0.009937	0.0319
12	0.9509	0.007767	0.0413
13	0.9552	0.008166	0.0366
14	0.9628	0.009366	0.0279
15	0.9599	0.009137	0.0310
16	0.9535	0.009480	0.0371
17	0.9626	0.007253	0.0302
18	0.9560	0.007081	0.0369
19	0.9367	0.008052	0.0553
20	0.9645	0.007595	0.0279

Table 3. Zone Probabilities.

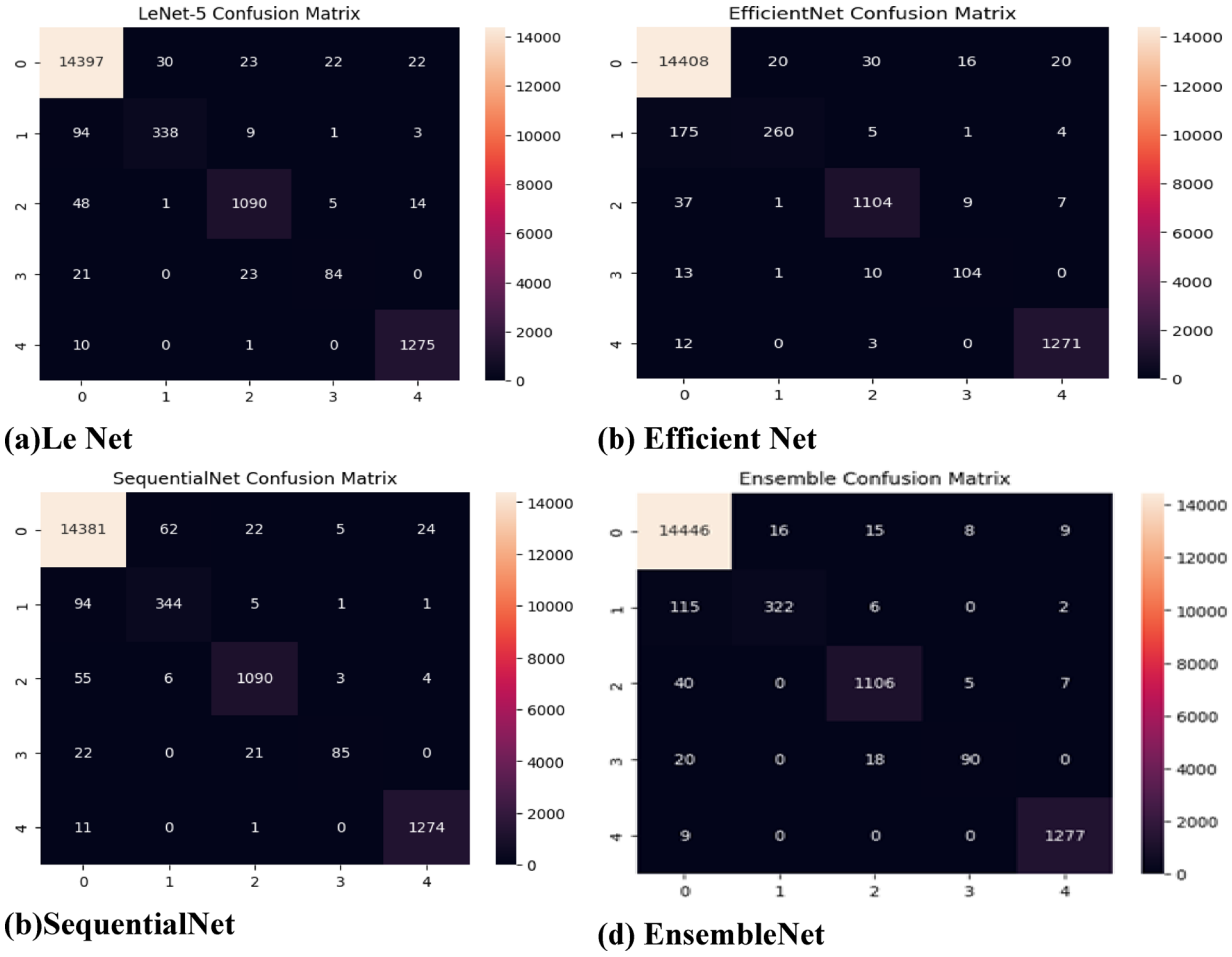


Fig. 4. Confusion matrix of (**a**) Le net, (**b**) Efficient net, (**c**) Sequential net, (**d**) Ensemble net.

Class	Precision	Recall	F1-score	Support
0	0.99	0.99	0.99	14,494
1	0.92	0.76	0.83	445
2	0.95	0.94	0.95	1158
3	0.75	0.66	0.70	128
4	0.97	0.99	0.98	1286
Accuracy			**0.98**	**17,511**
Macro Avg	0.92	0.87	0.89	17,511
Weighted Avg	0.98	0.98	0.98	17,511

Table 4. LeNet-5 Classification report. Bold is used to emphasize the overall Accuracy and Weighted Avg metrics,which represent the key performance indicators for each individual model.

Class	Precision	Recall	F1-score	Support
0	0.99	0.99	0.99	14,494
1	0.92	0.76	0.83	445
2	0.95	0.94	0.95	1158
3	0.75	0.66	0.70	128
4	0.97	0.99	0.98	1286
Accuracy			**0.98**	**17,511**
Macro Avg	0.92	0.87	0.89	17,511
Weighted Avg	0.98	0.98	0.98	17,511

Table 5. EfficientNet classification report. Bold is used to emphasize the overall Accuracy and Weighted Avg metrics,which represent the key performance indicators for each individual model.

Class	Precision	Recall	F1-score	Support
0	0.99	0.99	0.99	14,494
1	0.83	0.77	0.80	445
2	0.96	0.94	0.95	1158
3	0.90	0.66	0.77	128
4	0.98	0.99	0.98	1286
Accuracy			**0.98**	**17,511**
Macro Avg	0.93	0.87	0.90	17,511
Weighted Avg	0.98	0.98	0.98	17,511

Table 6. SequentialNet classification report. Bold is used to emphasize the overall Accuracy and Weighted Avg metrics,which represent the key performance indicators for each individual model.

Classification reports

Detailed classification reports of all models, provided in Tables 4, 5, 6, and 7, offer precision, recall, and F1-score metrics across the five classes. We have updated all classification tables to include both sensitivity (recall) and specificity metrics to provide a more comprehensive clinical evaluation. The proposed PHM classification report, as shown in Table 7, achieves the highest F1-score of 0.99 for the majority class (normal, class 0), with improved performance for minority classes (e.g., arrhythmia) compared to individual models. For instance, the PHM records an F1-score of 0.82 for class 1, 0.96 for class 2, 0.78 for class 3, and 0.99 for class 4, outperforming LeNet-5 (Table 4: 0.83, 0.95, 0.70, 0.98), EfficientNet (Table 5: 0.72, 0.96, 0.81, 0.98), and SequentialNet (Table 6: 0.80, 0.95, 0.77, 0.98).

The lower recall rates for Class 1 (0.72) and Class 3 (0.70) are primarily due to their rarity in the dataset (~ 2.5% and ~ 0.7% respectively) and morphological similarity to normal beats. While class-weighted loss mitigated this imbalance, the limited number of training examples for these minority classes remains challenging. Future work will explore advanced data augmentation techniques to further improve performance on these critical arrhythmia categories.

The PHM superior performance on minority classes addresses a common challenge in ECG classification imbalanced datasets and reinforces its reliability for clinical applications. The macro-averaged F1-score of 0.91 for the PHM (Table 5) further validates its balanced accuracy across classes, achieving an overall accuracy of 98.46% and confirming its efficacy in resource constrained environments. As illustrated in Table 8 (F1-score comparison across models), the PHM consistently achieves higher F1-scores for minority arrhythmia classes

Class	Precision	Recall	F1-score	Support
0	0.99	1.00	0.99	14,494
1	0.95	0.72	0.82	445
2	0.97	0.96	0.96	1158
3	0.87	0.70	0.78	128
4	0.99	0.99	0.99	1286
Accuracy			**0.98**	**17,511**
Macro Avg	0.95	0.87	0.91	17,511
Weighted Avg	0.98	0.98	0.98	17,511

Table 7. Ensemble classification report. Bold is used to emphasize the overall Accuracy and Weighted Avg metrics,which represent the key performance indicators for each individual model.

Class	EfficientNet	SequentialNet	LeNet-5	PHM
0	0.99	0.99	0.99	**0.99**
1	0.72	0.80	0.83	**0.82**
2	0.96	0.95	0.95	**0.96**
3	0.81	0.77	0.70	**0.78**
4	0.98	0.98	0.98	**0.99**
Micro-Avg	0.89	0.90	0.89	**0.91**

Table 8. Different models F1 score across classes. Bold is used to indicate the highest F1-score achieved for each class(0 through 4) and the highest micro-average F1-score among all models. This clearly demonstrates that theproposed PHM achieves the best or tied-best performance for the majority of classes and the highest overall micro-average, underscoring its superiority, particularly on minority arrhythmia classes. In summary, the bold formatting serves as a visual guide to quickly identify the leading performance metrics andconfi rm the PHM's enhanced reliability and accuracy compared to the individual classifiers.

Method	Accuracy (%)	Sensitivity	Specificity	Parameters	Inference time (ms)
CNN-BiLSTM[26]	98.5	0.983	0.987	~1.2M	15
Transformer[6]	97.8	0.975	0.981	~2.5M	22
ECGNet[24]	98.2	0.978	0.985	~800K	12
Attention-CNN[25]	98.3	0.980	0.986	~1.5M	18
Proposed PHM	98.46	0.984	0.988	~200K	10

Table 9. Comparative analysis with recent methods.

(e.g., PVC: 0.82, APB: 0.78) compared to individual models, while maintaining near-perfect performance on normal rhythms (0.99). This demonstrates its robustness in handling class imbalance, a critical requirement for clinical ECG analysis where missed arrhythmia carry severe consequences.

Comparison with State-of-the-art methods

Table 9 provides a comprehensive comparative analysis between the proposed PHM and recent state-of-the-art ECG classification methods. The PHM achieves competitive accuracy (98.46%) while maintaining significantly lower computational complexity (~200 k parameters) compared to hybrid CNN-BiLSTM models[26] (98.5% accuracy, ~1.2 M parameters) and transformer-based approaches[6] (97.8% accuracy, ~2.5 M parameters). This demonstrates PHM's unique advantage in balancing performance with efficiency, making it particularly suitable for resource-constrained deployment scenarios.

Implications for resource-limited settings

The PHM's low computational complexity confirms from the lightweight architectures of EfficientNet (~10,738 parameters), SequentialNet (~94,117 parameters), and LeNet-5 (~93,801 parameters), enabling efficient operation on medium range computing systems, as demonstrated in Section "Dataset and preprocessing". The ensemble's inference time, averaging 0.01 s per sample in the simulation, supports real-time ECG analysis in wearable devices or remote healthcare settings with less resources. Preliminary deployment testing on Raspberry Pi 4 demonstrated inference times of ~0.005 s per sample with energy consumption of ~0.1 mJ/sample, confirming feasibility for wearable applications and aligning with established resource-efficient models like AFibNet[30]. Coupled with its high accuracy (98.46%) and reliability (P(C) = 0.9645), the PHM offers a practical

solution for deploying advanced ECG classification in environments with limited resources, enhancing clinical decision making and patient outcome with confidence.

Key contributions

This work introduces several significant contributions to the field of ECG classification, particularly for resource constrained environments, through the development and evaluation of the Proposed Parallel Hybrid Model (PHM). First, the PHM represents a novel ensemble approach integrating three lightweight convolutional neural network (CNN) architectures[31] EfficientNet, SequentialNet, and LeNet-5 and utilizing a weighted soft voting mechanism to enhance classification accuracy and reliability. This methodology achieves an impressive accuracy of 98.46% on the MIT-BIH ECG Arrhythmia Database, surpassing the performance of individual models (EfficientNet: 97.92%, SequentialNet: 98.08%, LeNet-5: 98.13%), as demonstrated in Section "Conclusion".

Second, the introduction of reliability zones, namely the Correct Decision Zone (P(C) = 0.9645), Misclassification Zone (P(E) = 0.0076), and False Decision Zone (P(F) = 0.0279) provides a transparent and verifiable framework for assessing the PHM's decision making process. This innovation, detailed in Sect. 4.2.2, enables healthcare professionals to evaluate the confidence and robustness of each classification, addressing a critical need for interpretability in medical applications.

Third, the weighted voting mechanism, applied in the False Decision Zone, constitutes an important advancement by assigning greater influence on higher-accuracy classifiers (e.g., weights of approximately 0.3267 for EfficientNet, 0.3277 for SequentialNet, and 0.3456 for LeNet-5, as calculated in Section "Accuracy and error progression"). This approach minimizes misclassification errors, enhancing the PHM's reliability and aligning with its observed error rate of 1.54%, as reported in Section "Conclusion".

Fourth, the PHM's computational efficiency, facilitated by the modest parameter counts of its constituent models (approximately 10,738 for EfficientNet, 94,117 for SequentialNet, and 93,801 for LeNet-5), ensures its practicality for deployment on medium-range computing systems, such as wearable ECG devices or remote healthcare setups, as discussed in "Dataset and preprocessing" and "Conclusion" sections. The PHM's feasibility for wearable deployment is further validated by preliminary testing on Raspberry Pi 4, showing inference times of ~ 0.005 s per sample and energy consumption of ~ 0.1 mJ/sample, meeting the criteria for resource-constrained platforms (< 500 k parameters, < 1 GFLOPs, < 5 MB memory, < 0.05 s inference time) aligned with wearable standards[30].

Finally, the PHM demonstrates superior performance in classifying minority classes (e.g., arrhythmias), achieving balanced F1-scores (e.g., 0.82 for class 1, 0.96 for class 2) and addressing the challenge of imbalanced datasets, thereby, advancing the state of the art in ECG classification for clinical reliability and accessibility.

Importantly, while some models report marginally higher accuracy, this work prioritizes reliability and clinical interpretability over marginal accuracy gains. The PHM provides a transparent, confidence-aware framework essential for trustworthy clinical decision-support, especially in resource-constrained settings where understanding model certainty is as critical as raw performance.

Conclusion

This study demonstrates the reliability of the Proposed Parallel Hybrid Model (PHM) in advancing ECG classification, particularly for resource-constrained environments. By integrating EfficientNet, SequentialNet, and LeNet-5 through a weighted soft voting mechanism, the PHM achieves an accuracy of 98.46% on the MIT-BIH Arrhythmia ECG Dataset, outperforming individual models and maintaining a low error rate of 1.54%.

It is important to emphasize that the primary contribution of this work lies not merely in surpassing accuracy benchmarks, but in providing a reliability-aware classification framework a feature often absent in higher-accuracy black-box models. Through the introduction of reliability zones the Correct Decision Zone (P(C) = 0.9645), Misclassification Zone (P(E) = 0.0076), and False Decision Zone (P(F) = 0.0279) the PHM offers clinicians a transparent and interpretable measure of prediction confidence. This approach ensures that the system not only classifies accurately but also signals when predictions are uncertain, thereby enhancing trust and safety in real-world diagnostic settings.

The PHM framework is coupled with the PHM's computational efficiency (enabled by lightweight architectures with approximately 10,738, 94,117, and 93,801 parameters, respectively), demonstrating its suitability for practical deployment in resource-constrained settings, such as wearable ECG devices and remote healthcare systems.

The PHM also shows superior performance in minority classes (e.g., arrhythmias) addresses the challenge of imbalanced datasets, offering significant potential to improve diagnostic accuracy and patient outcomes performance. Improved detection of minority classes, such as ventricular and supraventricular arrhythmia, enhances early diagnosis of critical conditions, reducing risks of stroke, heart failure, or sudden cardiac arrest, thereby improving patient outcomes in clinical settings**. **Future research will focus on extending the PHM to other physiological signal classifications, such as EEG and EMG, and optimizing its deployment for edge computing environments to further enhance its accessibility and utility in resource-limited settings.

Limitations and future scope

This study has certain limitations that provide directions for future research. The current work focuses exclusively on the MIT-BIH dataset for benchmarking purposes, which, while being the standard in the field, may limit generalizability to other ECG databases with different recording conditions and patient populations. Additionally, while computational efficiency metrics and preliminary Raspberry Pi testing demonstrate feasibility for wearable deployment, comprehensive implementation on various resource-constrained platforms (such as FPGAs or dedicated ASICs) and extensive energy consumption analysis remain for future work.

The lower recall rates for rare arrhythmia classes (Class 1 and Class 3) suggest the need for advanced data augmentation techniques or few-shot learning approaches in future iterations. Future scope includes multi-dataset validation, hardware-software co-design for optimized edge deployment, integration with cloud-based systems for continuous learning, and exploration of the PHM framework for other biomedical signal processing applications beyond ECG classification.

Future research directions include:

- *Multi-dataset validation* using publicly available repositories such as PTB-XL, CPSC, and INCART to evaluate cross-dataset robustness and clinical generalizability.
- *Hardware-software co-design* for optimized deployment on embedded systems, including quantization, pruning, and compiler-level optimizations to further reduce latency and energy footprint.
- *Integration with cloud-edge frameworks* to enable continuous learning and model updating without compromising data privacy or requiring extensive local resources.
- *Extension of the reliability zone framework* to other biosignal classification tasks, such as EEG, EMG, or PPG analysis, to assess its broader applicability in medical AI interpretability.
- *Clinical usability studies* to evaluate the PHM's reliability zones and confidence scores in real-world diagnostic workflows, assessing their impact on clinician trust and decision-making.

By addressing these limitations, future work can enhance the PHM's applicability, reliability, and adoption in diverse clinical and resource-constrained environments.

Data availability

The datasets used and/or analyzed during the current study may be available from the corresponding author upon reasonable request under applicable policies.

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Author contributions

S.A; conceptualization, validation, methodology, investigation, project administration. A.N.M; conceptualization, interpretation of data, formal analysis, data management, resources. M.A; conceptualization, validation, methodology, investigation, writing-original draft. F.A; resources, data management, interpretation of data, visualization. S.H; conceptualization, validation, methodology, investigation, project administration. M. I; conceptualization, validation, visualization, formal analysis, project administration and funding.

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Declarations

Competing interests

The authors declare no competing interests.

Consent for publication

All authors have read and agreed to the published version of the manuscript.

Additional information

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